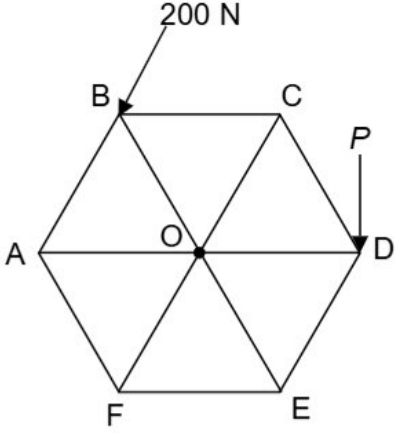
4 A merry-go-round is in the form of a regular hexagon ABCDEFA. It is free to rotate about a vertical axis through centre O as shown in the figure.



A child applied a 200 N force at B in the direction from B to A. Another child applied a force of *P* at D, and *P* is perpendicular to OD.

If the merry-go-round does not rotate, what is the magnitude of force *P*?

A 100 N **B** 170 N **C** 200 N **D** 230 N